Billy LaPlant

ADVANCED MANUFACTURING PLANNING ENGINEER

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Profile

- Results-driven engineer with hands-on experience in troubleshooting and building. Demonstrates strong communication skills and a passion for achieving team objectives. Proven track record of innovative problem-solving, with a focus on improving processes and reducing cost.

Experience

Subaru of Indiana Automotive, Inc. | Advanced Manufacturing Planning Engineer **Jan. 2025 – Current**

- Mentored a summer intern and newly hired engineer, assisted with tasks and gave advice to foster their technical and professional skills.
- Created a business proposal to revamp production lines, including cost analysis, building size, production volume, utility requirements, and stormwater plans.
- Formulated a business proposal for an energy efficient paint shop including energy requirements, carbon reduction technologies, staffing requirements, process automation specifications, and 2D layouts for multiple jobs per hour (JPH) scenarios.
- Investigated potential CO_2 reduction activities, including solar, nuclear, wind, and cogeneration.
- Completed vehicle studies for two production lines including cost estimation, construction and commissioning schedules, and takt time calculations.
- Analyzed new technologies and workforce reductions, calculating payback period and overall feasibility.
- Revamped plant technical standard revision process and wrote new training course.
- Led plant standards improvement efforts by engaging Engineering, Safety, Maintenance, IT and Production to collaborate on standards updates.
- Led studies on new technologies, including AI/RAG, smart factory, humanoid robots, energy savings and energy monitoring.
- Mediated a manager/coordinator communication conflict by promoting open dialogue, improving alignment and keeping the project on track.

Subaru of Indiana Automotive, Inc. | Paint Manufacturing Engineer **Jul. 2023 – Jan. 2025**

- Led a project replacing fifty conveyor sections completed one month ahead of schedule, by coordinating complex schedules with multiple departments.
- Managed a new model robot teaching project, collaborating with various departments and completing it one month sooner than previous efforts.
- Investigated new technologies and workforce reductions using CAD software and vendor proof of concepts.
- Led a project upgrading two robot applicators to accommodate a new model, reducing sealer overspray by 25% and increasing sealer coverage by 50%.
- Led a $400,000 conveyor inspection project of an entire paint shop to recognize wear patterns.
- Directed a project to install a new vision system and reactivate a dormant robot cell.

- Assisted in a project analyzing service parts, requiring daily trials for two months and clear communication with team members.
- Mentored a new engineer, facilitating his transition into the department and guiding him through his first project.

Subaru of Indiana Automotive, Inc. | CAP Engineer **Jun. 2022 – Jul. 2023**

- Completed a one-year rotational program, gaining hands-on experience in all departments of automotive manufacturing.
- Completed a 6-month rotation in paint manufacturing engineering, presented and completed three production-critical conveyor projects costing $600,000.
- Collaborated to inspect and upgrade sixty eyewash stations for compliance to ISO 45001 safety standards.

Index Corporation | Service Intern **May 2021 – Aug. 2021**

- Assembled and disassembled thirteen single-spindle and multi-spindle CNC machines.
- Troubleshot control panel drives on 2 CNC machines by replacing power cables and decreasing power supply.
- Replaced parts on IEMCA barloader and fixed errors on HMI to enable CNC machine operation.
- Aligned five barloaders to CNC machine using scope method and camera with software method.

Education

Purdue University, Master of Business & Technology **2025 – Current (Expected 2028)**

Purdue University, BS Mechatronics Engineering Technology **2018 – 2022**

Skills & Accomplishments

- Experience with CATIA, AutoCAD, SolidWorks, FANUC robots, Kawasaki robots, Mitsubishi PLC, Allen Bradley PLC
- Experience with PLC, G-Code, Java, C, C#, Visual Basic, and Python programming languages
- Eagle Scout